

SKIN CANCER RISK INTELLIGENCE & PATIENT DIAGNOSIS OPTIMIZATION

A Data-Driven Analysis of Patient Risk, Clinical Patterns,
and Preventive Healthcare Strategy

PREPARED FOR DERMALIFE ONCOLOGY & RESEARCH INSTITUTE (DORI)

Marian Ohwerhi | Senior Healthcare Data Analyst

Business Overview

Who We Are: A specialized healthcare organization dedicated to combating skin cancer through clinical research, advanced diagnostics, and population health analytics.

Our Mission



Reduce skin cancer mortality rates



Improve early-stage diagnosis success



Deliver personalized risk assessments



Support evidence-based medical decisions



Increase awareness of key health risks

Scale of Operations: Manages patient records across multiple regions, continuously monitoring symptoms, lesion characteristics, hereditary risk, and environmental conditions.

Business Problem

Rising late-stage skin cancer diagnoses, high-risk lesion progression, and recurring patient complications — particularly among elderly and underserved populations.

Suspected Contributing Factors (Pre-Analysis)



Environmental
Conditions



Hereditary
History



Smoking & Alcohol
Consumption



Poor Sanitation
Access

Why SQL Analysis Was Required: Despite large volumes of patient data, DORI lacked a structured framework to identify high-risk patients or urgent-intervention populations.

Project Objectives & Methodology

Project Objectives

- Identify high-risk patients
- Uncover diagnosis trends
- Assess environmental and lifestyle risk factors
- Deliver actionable, evidence-based recommendations

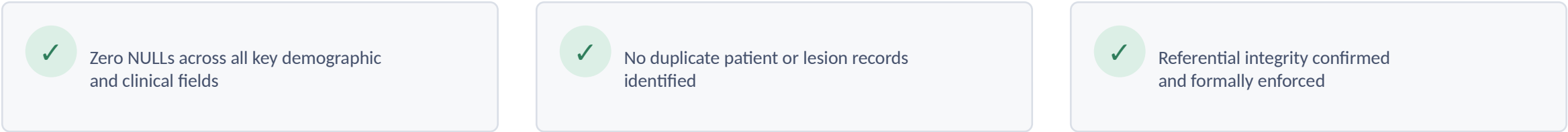
Methodology

- SQL-driven analysis using PostgreSQL
- Full data validation prior to analysis
- Referential integrity confirmed & formalized
- Findings cross-checked against known totals

Analytical Workflow



Validation Performed



Dataset Overview

1,088

Patients
Analyzed

1,088

Lesion Records
(1,007 distinct lesion IDs*)

6

Diagnosis
Categories

346

Malignant
Cases (32%)

Diagnosis Breakdown

Code	Type	Classification
ACK	Actinic Keratosis	Pre-cancerous
BCC	Basal Cell Carcinoma	Malignant
NEV	Nevus	Benign
SEK	Seborrheic Keratosis	Benign
SCC	Squamous Cell Carcinoma	Malignant
MEL	Melanoma	Malignant

Every patient record maps to exactly one lesion diagnosis
— verified through direct row-count validation.

**lesion_id is not globally unique (a data-dictionary discrepancy, confirmed by direct query). It was never used in any join or filter, so this does not affect any finding.*

Demographic Risk Analysis

REQUIRED FINDING • TASK 1, Q1

120

malignant diagnoses in the 60–74 age group — the highest of any age band by count

ADDITIONAL ANALYSIS • Malignancy Rate by Age Group

43.9%	33.7%	32.9%	20.3%	3.5%
75+	45–59	60–74	30–44	Under 30

Malignancy Rate by Gender

Female	Male
43.4%	26.0%

Previous Skin Cancer History

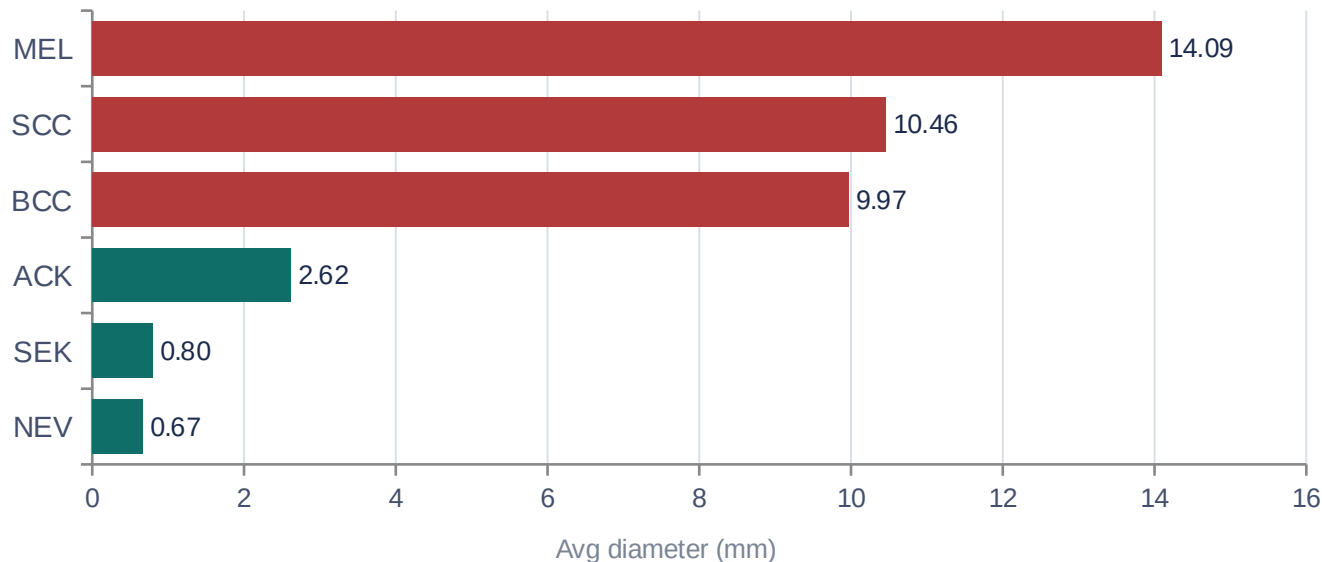
Documented history	No history
70.1%	21.9%

Required Task 1, Q2 — gender × diagnosis breakdown (all 6 diagnosis categories) is documented in full in the appendix.

Key Insight: Previous skin cancer history showed the strongest observed association with malignancy among the demographic factors examined — patients with a documented history make up roughly a fifth of the population but account for a disproportionate share of malignant cases.

Clinical & Lesion Analysis

Average Lesion Diameter by Diagnosis Type (Task 2, Q5)



MEL recorded the highest average lesion diameter at 14.09, followed by SCC at 10.46 and BCC at 9.97.

50.8%

malignancy rate among growing lesions, vs 15.1% for stable lesions (Additional Analysis 4) — growing lesions showed approximately 3.4x the observed malignancy rate of non-growing lesions.

100%

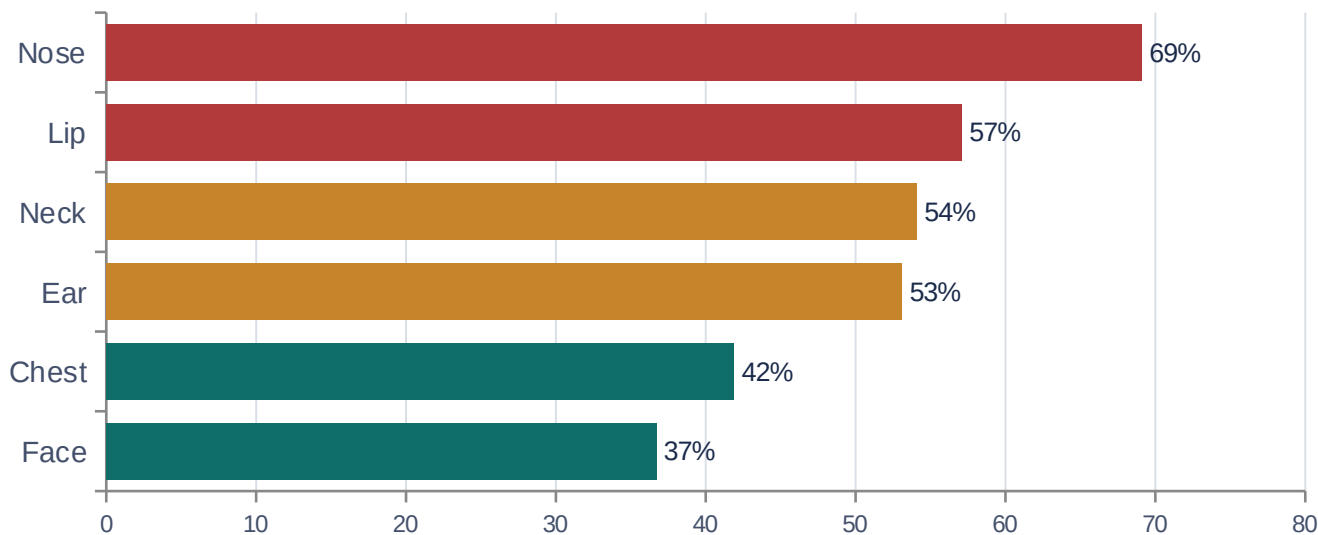
of lesions with malignant diagnoses were recorded as biopsied, vs 15.1% of non-malignant lesions (Additional Analysis 6).

Key Insight: Lesion growth and diameter both stand out as clinical warning signs; melanoma in particular presents with a distinctly larger average size than other diagnoses in this dataset.

Environmental Healthcare Analysis

ASSOCIATION, NOT CAUSATION — patients with better sanitation access show higher observed malignancy rates (~63% vs ~20%). This counterintuitive association may reflect differences in healthcare access, screening, or diagnostic follow-up, and warrants further investigation.

Malignancy Rate by Body Region — Top 6 (Additional Analysis 5)



124

biopsed lesions on the Face — the highest of any body region (Task 3, Q4).

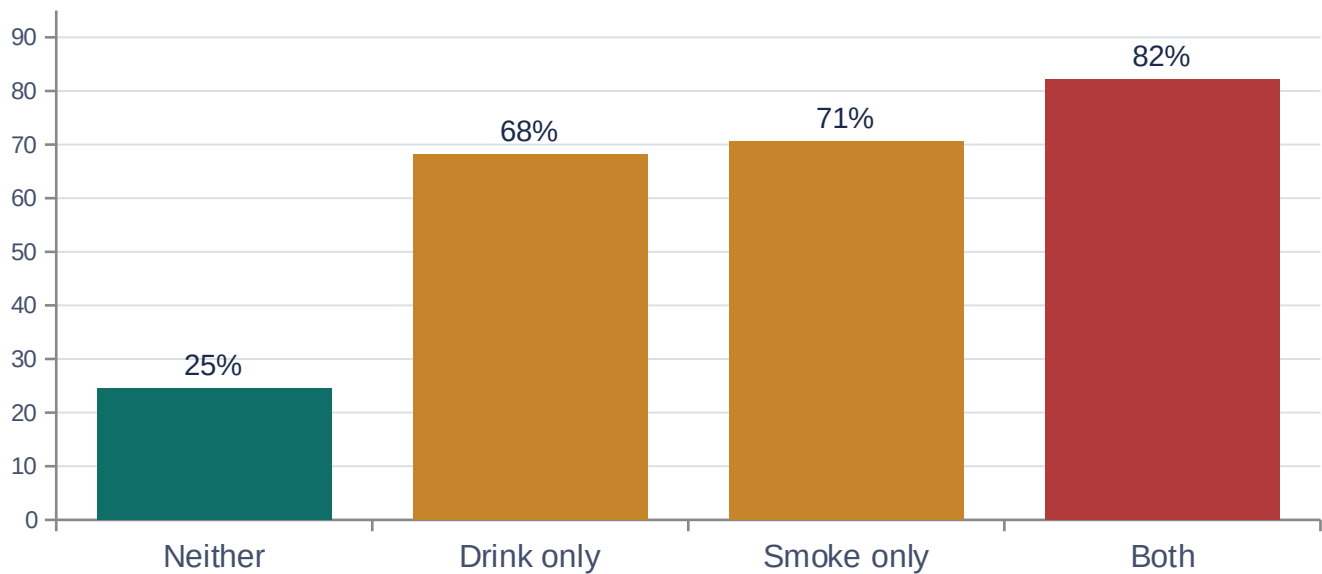
Sanitation Access vs. Malignancy Rate (Task 3, Q5)

Access	No Access	Has Access
Piped Water	19.3%	63.7%
Sewage System	21.5%	62.6%

Key Insight: Both utility measures move together (~63% vs ~20%). This may reflect a healthcare-access confound — patients with better infrastructure may be screened and diagnosed more consistently — but this dataset does not establish the underlying explanation. This warrants further investigation, not a causal conclusion.

Lifestyle & Behavioral Analysis

Malignancy Rate by Lifestyle Group — Four Mutually Exclusive Groups (Task 4, Q6)



REQUIRED • TASK 4, Q5

82.1%

smoke and drink

30.5%

all other patients

45.2%

of smokers also drink regularly (Task 4, Q4) — clustering well above the drinking rate in the wider patient population.

Key Insight: Patients who both smoke and drink showed the highest observed malignancy rate in this analysis (82.1%), well above the mutually exclusive smoke-only (70.6%) and drink-only (68.2%) groups, and nearly triple the rate among patients with neither habit (24.5%). This is an observed pattern in this dataset, not evidence that the two behaviors compound risk causally.

Executive Summary

Key Findings Across the Analysis

Finding	Observed Rate	Comparison
Smoking + Drinking Combined	82.1%	vs. 30.5% all other patients
Previous Skin Cancer History	70.1%	vs. 21.9% no prior history
Smoking Only (mutually exclusive)	70.6%	vs. 24.5% neither habit
Drinking Only (mutually exclusive)	68.2%	vs. 24.5% neither habit
Growing Lesions	50.8%	vs. 15.1% stable lesions
Age 75+ (highest rate)	43.9%	vs. 32.9% age 60–74
Nose (highest-rate body region)	69.1%	vs. 36.7% Face

These findings come from different subgroup analyses across the dataset and are not a single formal risk ranking. Rates above reflect observed associations; prior skin cancer history and combined smoking/drinking show the strongest associations, but none of these findings establish causation.

Strategic Recommendations

Immediate Actions

- ▶ Flag patients with prior skin cancer history for expedited screening (70.1% observed malignancy rate)
- ▶ Flag patients who both smoke and drink as a highest-priority screening tier (82.1% observed rate, vs. 30.5% for other patients)
- ▶ Prioritize scrutiny of nose, lip, neck, and ear lesions regardless of size

Medium-Term Actions

- ▶ Build patient education on lesion growth as a clinical warning sign (50.8% vs. 15.1% malignancy rate)
- ▶ Expand smoking/alcohol screening into standard intake forms with risk flagging

Further Investigation

- ▶ Investigate the sanitation-access association further, given the likely healthcare-access confound
- ▶ Explore drivers of elevated malignancy rate in patients aged 75+

Conclusion



Project Success

DORI's suspicion-driven questions were converted into validated, evidence-based findings across demographic, clinical, environmental, and behavioral dimensions.



Business Value

A clear, actionable risk-prioritization framework — enabling targeted screening rather than uniform resource allocation.



Clinical Value

Confirmation that current biopsy practices already track well with malignancy findings — validating existing clinical judgment while identifying specific gaps to close.

“This analysis transforms DORI's data from a passive record into an active tool for early detection — turning suspicion into evidence, and evidence into action.”

— END OF 12-SLIDE LIVE PRESENTATION — Pages that follow are a supplementary SQL appendix, for reference only

REFERENCE MATERIAL — NOT PART OF THE LIVE PRESENTATION

Appendix: SQL Query Library

Every required capstone question, paired with its final PostgreSQL query and verified output — grouped by data-quality checks, the four analytical tasks, and the six Additional Analyses. Included for transparency and reproducibility.

The 12 slides you present to DORI end at “Conclusion.” Everything from here on is backup only.

Data Validation & Integrity Checks

Check	Result	Interpretation
Global lesion_id uniqueness (total vs. distinct)	1,088 vs 1,007	lesion_id itself is not globally unique — a data-dictionary discrepancy (see note below); never used in a join/filter
Duplicate patient_id in table2 (one lesion row per patient)	0 rows	Clean — confirms each of the 1,088 patients has exactly one lesion row (this check uses patient_id, not lesion_id)
Orphan patient_id in table2	0 rows	Clean — every lesion maps to a real patient
NULLs — table1 (age, gender, skin_cancer_history)	0, 0, 0	Clean — no missing demographic data
NULLs — table2 (diagnostic, region, diameters, grew, biopsed)	all 0	Clean — no missing clinical data
Duplicate patient_id in table1	0 rows	Primary-key integrity confirmed, not just assumed
Whitespace/casing — region (14 distinct) & diagnostic (6 distinct)	clean	No hidden duplicate categories

DATA INTEGRITY FINDING — lesion_id is not globally unique (this does not mean lesion records are duplicated)

1,088 lesion records map to only 1,007 distinct lesion_id values — a discrepancy with the data dictionary, which describes lesion_id as a unique identifier. This is separate from the patient_id check above: every patient still has exactly one lesion row (confirmed above), but the lesion_id column itself repeats across some of those rows. Investigated and confirmed: this is a data-dictionary/documentation issue, not a duplication of patient or lesion records. lesion_id was never used in any JOIN, WHERE, or GROUP BY across this analysis, so it does not affect any finding in this presentation.

```
SELECT COUNT(*) AS total_lesions,
       COUNT(DISTINCT lesion_id)
       AS unique_lesion_ids
FROM table2;
```

Task 1 — Patient Demographic Risk Analysis (1 of 4)

Q1

Which age group has the highest number of skin cancer (malignant) diagnoses?

```
SELECT
  CASE
    WHEN t1.age < 30 THEN 'Under 30'
    WHEN t1.age BETWEEN 30 AND 44 THEN '30-44'
    WHEN t1.age BETWEEN 45 AND 59 THEN '45-59'
    WHEN t1.age BETWEEN 60 AND 74 THEN '60-74'
    WHEN t1.age >= 75 THEN '75+'
    ELSE 'Unknown'
  END AS age_group,
  COUNT(*) AS malignant_diagnoses
FROM table1 t1
JOIN table2 t2 ON t1.patient_id = t2.patient_id
WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL')
GROUP BY age_group
ORDER BY malignant_diagnoses DESC;
```

Result: 60-74 = 120 (highest); 45-59 = 111; 75+ = 83; 30-44 = 30; Under 30 = 2

Task 1 — Patient Demographic Risk Analysis (2 of 4)

Q2

What is the distribution of diagnoses between male and female patients?

```
SELECT
  t1.gender,
  t2.diagnostic,
  COUNT(*) AS num_cases
FROM table1 t1
JOIN table2 t2 ON t1.patient_id = t2.patient_id
GROUP BY t1.gender, t2.diagnostic
ORDER BY t1.gender, num_cases DESC;
```

Result: Female: ACK 135, BCC 122, NEV 42, SEK 28, SCC 25, MEL 10 | Male: ACK 326, BCC 151, SEK 109, MEL 102, SCC 31, NEV 7

Task 1 — Patient Demographic Risk Analysis (3 of 4)

Q3

Which body regions record the highest number of patients with malignant diagnoses?

```
SELECT
  region,
  COUNT(*) AS malignant_cases
FROM table2
WHERE diagnostic IN ('BCC', 'SCC', 'MEL')
GROUP BY region
ORDER BY malignant_cases DESC;
```

Result: Face leads (top region); full ranking across all 14 regions is in the query output

Task 1 — Patient Demographic Risk Analysis (4 of 4)

Q4

How many patients have a previous history of skin cancer?

```
SELECT  
    COUNT(*) AS patients_with_skin_cancer_history  
FROM table1  
WHERE skin_cancer_history = true;
```

Result: 224 patients (20.6% of the 1,088-patient population)

Task 2 — Lesion Growth & Diagnosis Analysis (1 of 5)

Q1

Which diagnosis category appears most frequently?

```
SELECT
  diagnostic,
  COUNT(*) AS num_lesions
FROM table2
GROUP BY diagnostic
ORDER BY num_lesions DESC;
```

Result: ACK is most frequent (461 of 1,088 lesions, 42.4%)

Task 2 — Lesion Growth & Diagnosis Analysis (2 of 5)

Q2

How many lesions were reported as growing over time?

```
SELECT  
  COUNT(*) AS growing_lesions  
FROM table2  
WHERE grew = true;
```

Result: 510 lesions were reported as growing

Task 2 — Lesion Growth & Diagnosis Analysis (3 of 5)

Q3

Which symptoms are most commonly associated with lesions?

```
SELECT 'itch' AS symptom, COUNT(*) AS num_lesions FROM table2 WHERE itch = true
UNION ALL
SELECT 'hurt', COUNT(*) FROM table2 WHERE hurt = true
UNION ALL
SELECT 'changed', COUNT(*) FROM table2 WHERE changed = true
UNION ALL
SELECT 'bleed', COUNT(*) FROM table2 WHERE bleed = true
UNION ALL
SELECT 'elevation', COUNT(*) FROM table2 WHERE elevation = true
ORDER BY num_lesions DESC;
```

Result: **Itch (670)** and **Elevation (611)** are the most common symptoms

Note: Symptoms are stored as separate boolean columns, so UNION ALL stacks five independent counts into one ranked list — GROUP BY alone could not do this.

Task 2 — Lesion Growth & Diagnosis Analysis (4 of 5)

Q4

How many lesions were biopsed before diagnosis confirmation?

```
SELECT  
    COUNT(*) AS biopsed_lesions  
FROM table2  
WHERE biopsed = true;
```

Result: 458 lesions were recorded as biopsed

Task 2 — Lesion Growth & Diagnosis Analysis (5 of 5)

Q5

Which diagnosis type has the highest average lesion diameter?

```
SELECT
  diagnostic,
  ROUND(AVG((diameter_1 + diameter_2) / 2.0)::numeric, 2)
  AS avg_lesion_diameter
FROM table2
GROUP BY diagnostic
ORDER BY avg_lesion_diameter DESC;
```

Result: MEL = 14.09 (highest); SCC 10.46; BCC 9.97; ACK 2.62; SEK 0.80; NEV 0.67

Note: Added to the master SQL as it is a required question in the capstone brief. AVG is cast to ::numeric before ROUND, since PostgreSQL rejects ROUND on a double-precision value.

Task 3 — Environmental Healthcare Analysis (1 of 5)

Q1

Which body region has the highest number of diagnosed cases (all diagnosis types)?

```
SELECT
  region,
  COUNT(*) AS total_cases
FROM table2
GROUP BY region
ORDER BY total_cases DESC;
```

Result: Face has the highest all-diagnosis case count

Task 3 — Environmental Healthcare Analysis (2 of 5)

Q2

How many patients lack access to piped water?

```
SELECT  
    COUNT(*) AS patients_without_piped_water  
FROM table1  
WHERE has_piped_water = false;
```

Result: 782 patients lack piped water access

Task 3 — Environmental Healthcare Analysis (3 of 5)

Q3

How many patients do not have access to sewage systems?

```
SELECT  
    COUNT(*) AS patients_without_sewage_system  
FROM table1  
WHERE has_sewage_system = false;
```

Result: 815 patients lack sewage system access

Task 3 — Environmental Healthcare Analysis (4 of 5)

Q4

Which body regions report the highest number of biopsed lesions?

```
SELECT
  region,
  COUNT(*) AS biopsed_count
FROM table2
WHERE biopsed = true
GROUP BY region
ORDER BY biopsed_count DESC;
```

Result: Face = 124 (highest); Chest 63; Nose 54; Back 45; Forearm 44

Task 3 — Environmental Healthcare Analysis (5 of 5)

Q5

Is there a relationship between poor sanitation access and severe diagnosis outcomes?

First branch of the UNION ALL:

```
SELECT
  'Piped Water' AS sanitation_type,
  t1.has_piped_water AS has_access,
  COUNT(*) AS total_patients,
  COUNT(*) FILTER (WHERE t2.diagnostic
    IN ('BCC', 'SCC', 'MEL'))
    AS malignant_cases,
  ROUND(100.0 * COUNT(*) FILTER
    (WHERE t2.diagnostic IN
    ('BCC', 'SCC', 'MEL')) / COUNT(*), 1)
    AS malignancy_rate_pct
FROM table1 t1
JOIN table2 t2
  ON t1.patient_id = t2.patient_id
GROUP BY t1.has_piped_water
```

ORDER BY sanitation_type, has_access; -- applies to the combined UNION ALL result

Result: Piped water: 19.3% (no access) vs 63.7% (access). Sewage: 21.5% vs 62.6%

Second branch, combined with UNION ALL:

```
SELECT
  'Sewage System' AS sanitation_type,
  t1.has_sewage_system AS has_access,
  COUNT(*) AS total_patients,
  COUNT(*) FILTER (WHERE t2.diagnostic
    IN ('BCC', 'SCC', 'MEL'))
    AS malignant_cases,
  ROUND(100.0 * COUNT(*) FILTER
    (WHERE t2.diagnostic IN
    ('BCC', 'SCC', 'MEL')) / COUNT(*), 1)
    AS malignancy_rate_pct
FROM table1 t1
JOIN table2 t2
  ON t1.patient_id = t2.patient_id
GROUP BY t1.has_sewage_system
```

Note: Framed as an observed association — not causation. It may reflect differences in healthcare access, screening, or diagnostic follow-up, and warrants further investigation.

Task 4 — Lifestyle & Behavioural Risk Analysis (1 of 6)

Q1

How many patients are smokers?

```
SELECT  
    COUNT(*) AS num_smokers  
FROM table1  
WHERE smoke = true;
```

Result: 62 patients are smokers

Task 4 — Lifestyle & Behavioural Risk Analysis (2 of 6)

Q2

How many patients consume alcohol regularly?

```
SELECT  
  COUNT(*) AS num_drinkers  
FROM table1  
WHERE drink = true;
```

Result: 138 patients consume alcohol regularly

Task 4 — Lifestyle & Behavioural Risk Analysis (3 of 6)

Q3

Which diagnosis types are most common among smokers?

```
SELECT
  t2.diagnostic,
  COUNT(*) AS num_cases
FROM table1 t1
JOIN table2 t2 ON t1.patient_id = t2.patient_id
WHERE t1.smoke = true
GROUP BY t2.diagnostic
ORDER BY num_cases DESC;
```

Result: BCC most common among smokers (30 of 62 lesions)

Task 4 — Lifestyle & Behavioural Risk Analysis (4 of 6)

Q4

What percentage of smokers also consume alcohol?

```
SELECT
  COUNT(*) AS total_smokers,
  COUNT(*) FILTER (WHERE drink = true)
    AS smokers_who_also_drink,
  ROUND(100.0 * COUNT(*) FILTER (WHERE drink = true)
    / COUNT(*), 1) AS pct_smokers_who_drink
FROM table1
WHERE smoke = true;
```

Result: 45.2% of smokers (28 of 62) also drink

Task 4 — Lifestyle & Behavioural Risk Analysis (5 of 6)

Q5

Are patients who both smoke and drink more likely to have a malignant diagnosis?

```
SELECT
  CASE
    WHEN t1.smoke = true AND t1.drink = true THEN 'Smoke and drink'
    ELSE 'Other patients'
  END AS lifestyle_group,
  COUNT(*) AS total_patients,
  COUNT(*) FILTER (WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL')) AS malignant_cases,
  ROUND(100.0 * COUNT(*) FILTER (WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL'))
    / COUNT(*), 1) AS malignancy_rate_pct
FROM table1 t1
JOIN table2 t2 ON t1.patient_id = t2.patient_id
GROUP BY lifestyle_group
ORDER BY malignancy_rate_pct DESC;
```

Result: Smoke and drink: 82.1% (28 patients) vs. Other patients: 30.5% (1,060 patients)

Note: An observed association in this dataset, not proof of causation.

Task 4 — Lifestyle & Behavioural Risk Analysis (6 of 6)

Q6

Which lifestyle factor has the strongest relationship with severe diagnosis outcomes?

```
SELECT
  CASE
    WHEN t1.smoke = true AND t1.drink = true THEN 'Smoke and drink'
    WHEN t1.smoke = true AND t1.drink = false THEN 'Smoke only'
    WHEN t1.smoke = false AND t1.drink = true THEN 'Drink only'
    ELSE 'Neither'
  END AS lifestyle_group,
  COUNT(*) AS total_patients,
  COUNT(*) FILTER (WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL')) AS malignant_cases,
  ROUND(100.0 * COUNT(*) FILTER (WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL'))
    / COUNT(*), 1) AS malignancy_rate_pct
FROM table1 t1
JOIN table2 t2 ON t1.patient_id = t2.patient_id
GROUP BY lifestyle_group
ORDER BY malignancy_rate_pct DESC;
```

Result: Both = 82.1% (28); Smoke only = 70.6% (34); Drink only = 68.2% (110); Neither = 24.5% (916)

Additional Analysis (1 of 6)

1

Malignancy rate by age group

```
SELECT
  CASE
    WHEN t1.age < 30 THEN 'Under 30'
    WHEN t1.age BETWEEN 30 AND 44 THEN '30-44'
    WHEN t1.age BETWEEN 45 AND 59 THEN '45-59'
    WHEN t1.age BETWEEN 60 AND 74 THEN '60-74'
    WHEN t1.age >= 75 THEN '75+'
    ELSE 'Unknown'
  END AS age_group,
  COUNT(*) AS total_patients,
  COUNT(*) FILTER (WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL')) AS malignant_cases,
  ROUND(100.0 * COUNT(*) FILTER (WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL'))
    / COUNT(*), 1) AS malignancy_rate_pct
FROM table1 t1
JOIN table2 t2 ON t1.patient_id = t2.patient_id
GROUP BY age_group
ORDER BY malignancy_rate_pct DESC;
```

Result: 75+ = 43.9% (highest rate); 45-59 = 33.7%; 60-74 = 32.9%; 30-44 = 20.3%; Under 30 = 3.5%

Note: A rate view of Task 1, Q1, which answered COUNT rather than rate. 60-74 has the highest malignant count (120); 75+ has the highest malignancy rate (43.9%) — these answer different questions.

Additional Analysis (2 of 6)

2

Malignancy rate by gender

```
SELECT
  t1.gender,
  COUNT(*) AS total_patients,
  COUNT(*) FILTER (WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL')) AS malignant_cases,
  ROUND(100.0 * COUNT(*) FILTER (WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL'))
    / COUNT(*), 1) AS malignancy_rate_pct
FROM table1 t1
JOIN table2 t2 ON t1.patient_id = t2.patient_id
GROUP BY t1.gender
ORDER BY malignancy_rate_pct DESC;
```

Result: Female = 43.4% (362 patients, 157 malignant); Male = 26.0% (726 patients, 189 malignant)

Note: A rate view of Task 1, Q2, which answered a count × diagnostic breakdown.

Additional Analysis (3 of 6)

3

Previous skin cancer history and malignancy rate

```
SELECT
  t1.skin_cancer_history,
  COUNT(*) AS total_patients,
  COUNT(*) FILTER (WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL')) AS malignant_cases,
  ROUND(100.0 * COUNT(*) FILTER (WHERE t2.diagnostic IN ('BCC', 'SCC', 'MEL'))
    / COUNT(*), 1) AS malignancy_rate_pct
FROM table1 t1
JOIN table2 t2 ON t1.patient_id = t2.patient_id
GROUP BY t1.skin_cancer_history;
```

Result: Yes (224 patients, 20.6%) = 70.1% malignancy rate; No (864 patients, 79.4%) = 21.9%

Note: Explores Task 1, Q4, which answered a plain count of patients with prior history.

Additional Analysis (4 of 6)

4

Growing vs. non-growing lesions and malignancy rate

```
SELECT
    grew,
    COUNT(*) AS total_lesions,
    COUNT(*) FILTER (WHERE diagnostic IN ('BCC', 'SCC', 'MEL')) AS malignant_cases,
    ROUND(100.0 * COUNT(*) FILTER (WHERE diagnostic IN ('BCC', 'SCC', 'MEL'))
        / COUNT(*), 1) AS malignancy_rate_pct
FROM table2
GROUP BY grew;
```

Result: Growing = 50.8% (510 lesions, 259 malignant); Not growing = 15.1% (578 lesions, 87 malignant)

Note: A rate view of Task 2, Q2, which answered a plain count of growing lesions.

Additional Analysis (5 of 6)

5

Malignancy rate by body region

```
SELECT
    region,
    COUNT(*) AS total_lesions,
    COUNT(*) FILTER (WHERE diagnostic IN ('BCC', 'SCC', 'MEL')) AS malignant_cases,
    ROUND(100.0 * COUNT(*) FILTER (WHERE diagnostic IN ('BCC', 'SCC', 'MEL'))
        / COUNT(*), 1) AS malignancy_rate_pct
FROM table2
GROUP BY region
ORDER BY malignancy_rate_pct DESC;
```

Result: Nose 69.1%; Lip 57.1%; Neck 54.1%; Ear 53.1%; Chest 41.9%; Face 36.7% (top 6 of 14 regions)

Note: A deeper cut beyond Task 1, Q3 and Task 3, Q1, which both answered counts rather than rates.

Additional Analysis (6 of 6)

6

Biopsy rate by malignancy status

```
SELECT
  CASE
    WHEN diagnostic IN ('BCC', 'SCC', 'MEL') THEN 'Malignant'
    ELSE 'Non-malignant'
  END AS diagnosis_type,
  COUNT(*) AS total_lesions,
  COUNT(*) FILTER (WHERE biopsed = true) AS biopsed_count,
  ROUND(100.0 * COUNT(*) FILTER (WHERE biopsed = true) / COUNT(*), 1) AS biopsy_rate_pct
FROM table2
GROUP BY diagnosis_type;
```

Result: Malignant: 346 total, 346 biopsed, 100.0% rate; Non-malignant: 742 total, 112 biopsed, 15.1% rate

Note: Explores Task 2, Q4, which answered a plain count of biopsed lesions.